Notebook 3.3 - Visualization & Integration

Management Science - CEO Presentations That Wow!

Introduction

Welcome to your final CEO challenge at Bean Counter!

The Board Meeting Challenge

Tomorrow morning, you're presenting Bean Counter's annual performance to the board of directors and potential investors. You have:

- Massive datasets analyzed with NumPy
- Complex reports created with Pandas
- Incredible insights about the business

The Problem: Numbers alone don't convince boards. They need to SEE the story! A table with 1000 rows won't inspire investment, but a compelling chart will.

Your Solution: Data Visualization - turning your analyses into powerful visual stories that drive decisions and secure funding.

In this tutorial, you'll create the visualizations that will secure Bean Counter's future and cement your legacy as CEO.

> ⚠ How to Use This Tutorial
>
> Import matplotlib along with pandas and numpy. Remember, we work with `uv`, so just use `uv add matplotlib` in the terminal. Now you can create compelling charts for your board presentation!

> Start practicing generate AI
>
> The creation of plots is a perfect opportunity to start practicing generate AI. Usually you need a lot of code for nice visualizations and most of it is repetitive.

Section 1 - Bar Charts for Performance Comparison

As CEO, you often need to compare performance across stores, products, or time periods. Bar charts are perfect for this.

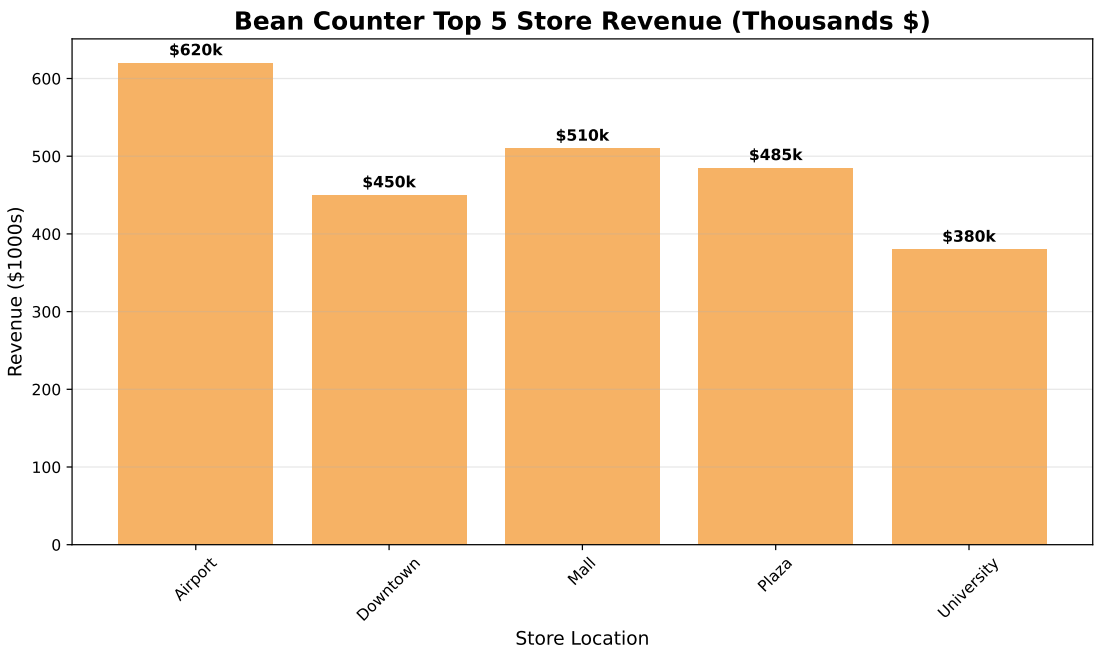
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
# Top 5 store performance
stores = ['Airport', 'Downtown', 'Mall', 'Plaza', 'University']
revenue = [620, 450, 510, 485, 380]

# Create bar chart
plt.figure(figsize=(10, 6))
plt.bar(stores, revenue, color='#F6B265')
plt.title('Bean Counter Top 5 Store Revenue (Thousands $)', fontsize=16,
fontweight='bold')
plt.xlabel('Store Location', fontsize=12)
plt.ylabel('Revenue ($1000s)', fontsize=12)
plt.grid(axis='y', alpha=0.3)
plt.xticks(rotation=45)

# Add value labels on bars
for i, v in enumerate(revenue):
    plt.text(i, v + 10, f'${v}k', ha='center', fontweight='bold')

plt.tight_layout()
plt.show()
```



Bar Charts Are Your Best Friend

As CEO, bar charts will be your most-used visualization:

- Compare stores, products, or departments
- Show rankings clearly
- Easy for board members to understand
- Perfect for “top 10” or “bottom 5” analyses

Exercise 1.1 - Product Performance Bar Chart

Create a bar chart showing Bean Counter’s product sales for the board meeting.

```
import pandas as pd
import matplotlib.pyplot as plt

# Product sales data
products_df = pd.DataFrame({
     'product': ['Latte', 'Espresso', 'Cappuccino', 'Americano', 'Mocha',
'Macchiato'],
    'units_sold': [3200, 4500, 2800, 2100, 1900, 1500]
})

# YOUR CODE BELOW
# 1. Sort products by units_sold (highest first)
products_sorted =

# 2. Create a bar chart
```

Section 2 - Line Plots for Trends Over Time

CEOs need to show growth and trends. Line plots are perfect for time series data.

```
import pandas as pd
import matplotlib.pyplot as plt

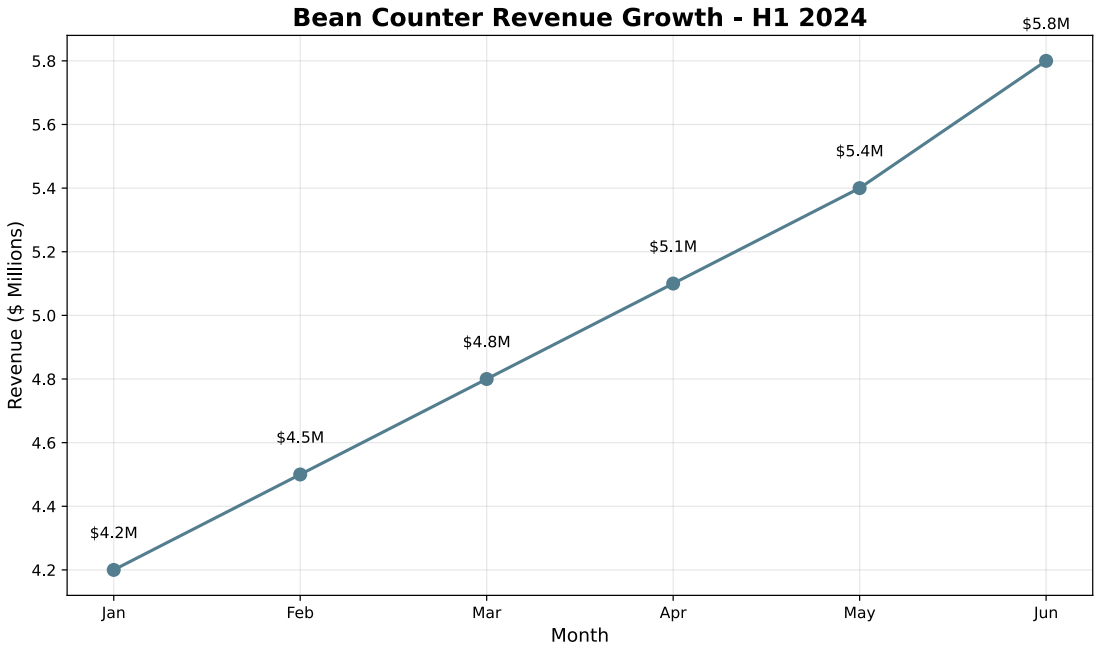
# Monthly revenue trend
months = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun']
revenue = [4.2, 4.5, 4.8, 5.1, 5.4, 5.8]

plt.figure(figsize=(10, 6))
plt.plot(months,    revenue,    marker='o',    linewidth=2,    markersize=8,
color='#537E8F')
plt.title('Bean   Counter   Revenue   Growth   -   H1   2024',   fontsize=16,
fontweight='bold')
plt.xlabel('Month', fontsize=12)
plt.ylabel('Revenue ($ Millions)', fontsize=12)
```

```
plt.grid(True, alpha=0.3)

# Add value labels
for i, v in enumerate(revenue):
    plt.text(i, v + 0.1, f'${v}M', ha='center')

plt.tight_layout()
plt.show()
```



Exercise 2.1 - Customer Growth Trend

Show the board Bean Counter's customer growth over the past 8 months.

```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np

# Customer data (in thousands)
months = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun', 'Jul', 'Aug']
customers = [145, 152, 158, 165, 174, 182, 195, 208]

# YOUR CODE BELOW
# Create a line plot showing customer growth
```

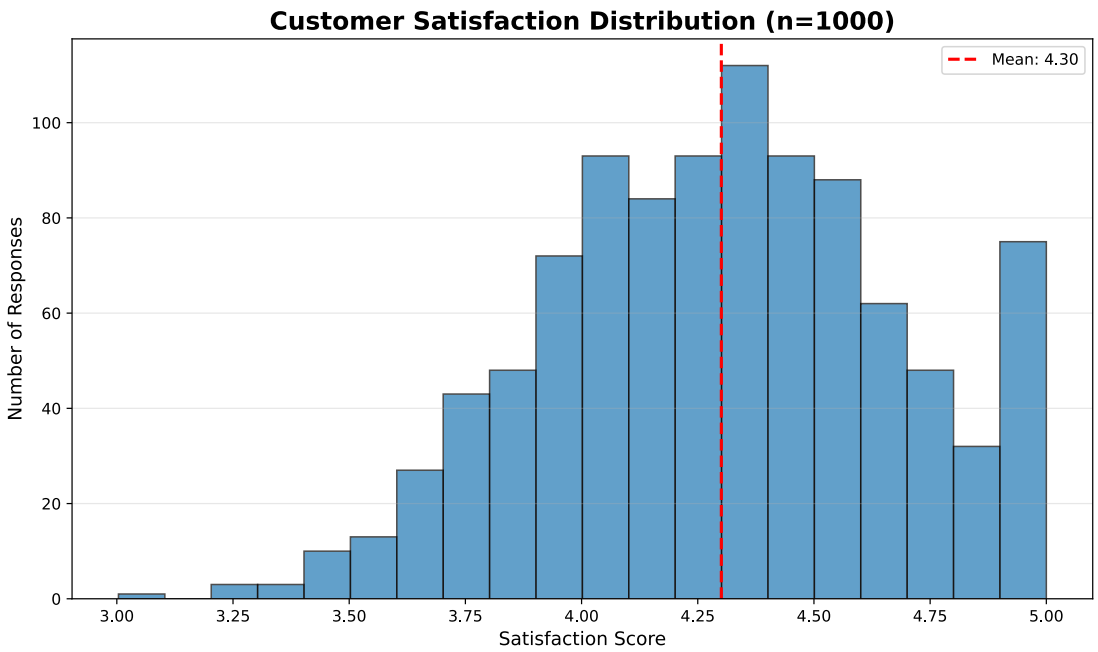
Section 3 - Histograms for Distribution Analysis

Show the board how metrics are distributed across your empire.

```
import numpy as np
import matplotlib.pyplot as plt

# Simulate customer satisfaction scores
np.random.seed(42)
satisfaction_scores = np.random.normal(4.3, 0.4, 1000)
satisfaction_scores = np.clip(satisfaction_scores, 1, 5)

plt.figure(figsize=(10, 6))
plt.hist(satisfaction_scores, bins=20, alpha=0.7, edgecolor='black')
plt.title('Customer Satisfaction Distribution (n=1000)', fontsize=16,
fontweight='bold')
plt.xlabel('Satisfaction Score', fontsize=12)
plt.ylabel('Number of Responses', fontsize=12)
plt.axvline(x=satisfaction_scores.mean(), color='red', linestyle='--',
linewidth=2,
           label=f'Mean: {satisfaction_scores.mean():.2f}')
plt.legend()
plt.grid(axis='y', alpha=0.3)
plt.tight_layout()
plt.show()
```



Exercise 3.1 - Store Efficiency Distribution

Show the board how store efficiency is distributed across all locations.

```
import numpy as np
import matplotlib.pyplot as plt
```

```
# Simulate efficiency scores for 50 stores
np.random.seed(100)
efficiency_scores = np.random.normal(75, 12, 50)  # Mean 75, std 12

# YOUR CODE BELOW
# Create histogram of efficiency distribution
```

Section 4 - Integrated Analysis with Pandas

Combine Pandas data manipulation with visualization for powerful insights.

```
import pandas as pd
import matplotlib.pyplot as plt

# Quarterly performance data
quarters_df = pd.DataFrame({
    'quarter': ['Q1', 'Q2', 'Q3', 'Q4'],
    'revenue': [4.2, 4.8, 5.3, 5.9],
    'profit': [1.3, 1.5, 1.8, 2.1],
    'customers': [145, 165, 188, 210]
})

# Create subplots for comprehensive view
fig, axes = plt.subplots(1, 3, figsize=(15, 5))

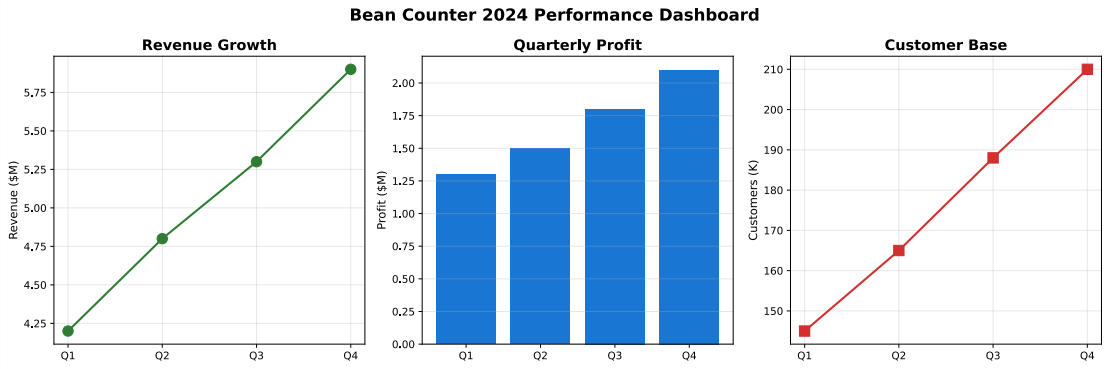
# Revenue trend
axes[0].plot(quarters_df['quarter'], quarters_df['revenue'],
            marker='o', linewidth=2, markersize=10, color='#2E7D32')
axes[0].set_title('Revenue Growth', fontsize=14, fontweight='bold')
axes[0].set_ylabel('Revenue ($M)', fontsize=11)
axes[0].grid(True, alpha=0.3)

# Profit bars
axes[1].bar(quarters_df['quarter'], quarters_df['profit'], color='#1976D2')
axes[1].set_title('Quarterly Profit', fontsize=14, fontweight='bold')
axes[1].set_ylabel('Profit ($M)', fontsize=11)
axes[1].grid(axis='y', alpha=0.3)

# Customer growth
axes[2].plot(quarters_df['quarter'], quarters_df['customers'],
            marker='s', linewidth=2, markersize=10, color='#D32F2F')
axes[2].set_title('Customer Base', fontsize=14, fontweight='bold')
axes[2].set_ylabel('Customers (K)', fontsize=11)
axes[2].grid(True, alpha=0.3)

plt.suptitle('Bean Counter 2024 Performance Dashboard', fontsize=16,
```

```
fontweight='bold')
plt.tight_layout()
plt.show()
```



Exercise 4.1 - Complete CEO Dashboard

Create an integrated dashboard combining NumPy simulation with Pandas analysis and visualization. Create revenue projections for the company:

- Start at $6M in January
- Grow approximately 2% each month
- Add random variation (±25%) to simulate market fluctuations

> **Tip**
>
> To create realistic monthly growth:
>
> - Start with base revenue of 6.0M
> - Each month, apply 2% growth: `revenue = previous_revenue * 1.02`
> - Add random variation: multiply by `np.random.uniform(0.75, 1.25)`

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Simulate next year's projections
np.random.seed(42)
months = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun',
          'Jul', 'Aug', 'Sep', 'Oct', 'Nov', 'Dec']

# YOUR CODE BELOW
# 1. Create revenue projections for the company
# Start with 6.0M, grow each month by 2% with random variation
revenue_projection = []
current_revenue = 6.0  # Starting revenue in millions
```

```
# Loop through 12 months


# 2. Create DataFrame with months and revenue


# 3. Calculate cumulative revenue


# 4. Create visualization with two subplots
```

Conclusion

Congratulations! You've completed your CEO journey at Bean Counter!

You've mastered:

- **Bar Charts** - Comparing performance across stores and products
- **Line Plots** - Showing growth trends over time
- **Histograms** - Understanding distributions of key metrics
- **Integrated Dashboards** - Combining multiple visualizations for comprehensive insights
- **NumPy + Pandas + Visualization** - The complete data science toolkit

Your Bean Counter CEO visualization toolkit enables you to:

- Present compelling stories to the board of directors
- Identify trends and patterns at a glance
- Compare performance across multiple dimensions
- Make data-driven decisions with confidence
- Communicate complex insights simply and effectively

Remember:

- Bar charts are best for comparisons
- Line plots show trends over time
- Histograms reveal distributions
- Always label your axes and add titles
- Use colors consistently across related charts
- Annotations can highlight key insights

Your Bean Counter Legacy:

From Barista Trainee to CEO, you've transformed Bean Counter into a data-driven enterprise.

- Standardized operations with functions
- Managed complex data with dictionaries
- Optimized decisions with sorting algorithms

- Processed big data with NumPy's speed
- Analyzed business metrics with Pandas
- Communicated insights with powerful visualizations

What's Next: Armed with these data science foundations, you're ready for the advanced algorithms in upcoming lectures! You'll tackle Monte Carlo simulations, forecasting, scheduling optimization, and more. The skills you've built here - from basic Python to advanced visualization - will be the foundation for solving complex management science problems!

Solutions

You will likely find solutions to most exercises online. However, I strongly encourage you to work on these exercises independently without searching explicitly for the exact answers to the exercises. Understanding someone else's solution is very different from developing your own. Use the lecture notes and try to solve the exercises on your own. This approach will significantly enhance your learning and problem-solving skills.

Remember, the goal is not just to complete the exercises, but to understand the concepts and improve your programming abilities. If you encounter difficulties, review the lecture materials, experiment with different approaches, and don't hesitate to ask for clarification during class discussions.

Bibliography